

OPTIMIZED INVENTORY MANAGEMENT USING A HYBRID MACHINE LEARNING MIXED-INTEGER LINEAR PROGRAMMING APPROACH.

Hailley Wanjiru Njunji

Supervised by: Dr. Dennis Kaburu

A proposal submitted to the Department of Information Technology in the School of Computing and Information Technology in partial fulfilment of the requirement for the award of the degree of Bachelor of Science in Information Technology in the Jomo Kenyatta University of Agriculture and Technology.

January, 2025

Declaration

This proposal/research project is my original work and has not been presented for a degree in any other University

.....

Signature

Hailley Wanjiru Njunji

.....

Date

This proposal/research project has been submitted for examination with my approval as University Supervisor

.....

Signature

Dr. Dennis Kaburu

.....

Date

Abstract

In many small to medium organizations inventory management is characterized by reactive, manual heuristics and ad hoc decision making particularly regarding Bulk to Break unit conversions. These practices often culminate in a significant end of month operational crisis, where excessive labour is required to reconcile stock levels and rectify procurement errors. Though functional, these methods leave room for risks like overstocking and stock outs which lead to lost revenue and service level failures. This research aims to mitigate existing inefficiencies by designing and assessing an integrated framework that combines Machine Learning (ML) with Mixed-Integer Linear Programming (MILP) for inventory optimization. Guided by the CRISP-DM methodology, the study employs an XGBoost regressor to capture fragmented demand patterns, generating stochastic forecasts that subsequently feed into an MILP model constructed using the PuLP library.

The proposed framework mathematically models inventory flow while rigorously incorporating constraints such as storage capacity, holding costs and fixed ordering charges. Using controlled discrete event simulations, the hybrid ML–MILP policy is evaluated against a conventional fixed interval manual approach. Anticipated results suggest that this mathematically structured method can substantially lower overall inventory costs, stabilize stock availability and reduce the need for labour intensive reconciliation processes. In doing so, the study offers a data driven alternative to heuristic inventory practices, thereby promoting greater operational efficiency and enhancing data reliability within SME packaging enterprises.

Table of Contents

Declaration.....	I
Abstarct.....	II
List of Tables	V
List of Figures	VI
Acronyms.....	VII
Definition of Terms.....	VIII
CHAPTER 1: INTRODUCTION	1
1.1. Background	1
1.2. Project Overview.....	2
1.3. Statement of the Problem	3
1.4. Proposed Solution	3
1.5. Objectives.....	4
1.5.1. General Objective	4
1.5.2. Specific Objectives	4
1.6. Research Questions	4
1.7. Justification	4
1.8. Proposed System Methodologies	5
1.9. Scope.....	6
CHAPTER 2: LITERATURE REVIEW	7
2.1. Introduction.....	7
2.2 Theoretical Review	7
2.2.1 Inventory Management Concepts	7
2.2.2 Inventory Control Policies	7
2.2.3 Optimization and MILP in Inventory Decision-Making	8
2.2.4 Why Hybrid ML + MILP (Predictive–Prescriptive Analytics)	8
2.3 Case Study Review	9
2.4 Integration and Architecture	9
2.5 Summary	10
2.6 Research Gaps.....	10
References.....	12
APPENDICES	13
APPENDIX A: PROJECT RESOURCES.....	13
APPENDIX B: PROJECT BUDGET	14

APPENDIX C: PROJECT SCHEDULE	14
APPENDIX D: GANTT CHART.....	14

List of Tables

Table 1: Budget	13
Table 2: Schedule	13

List of Figures

Figure 1	13
----------------	----

Acronyms

LP - Linear Programming

MILP - Mixed-Integer Linear Programming

ML – Machine Learning

ISO - International Organization for Standardization

SME - Small and Medium Enterprise

CRISP-DM - Cross-Industry Standard Process for Data Mining

Definition of Terms

Inventory Policy – This is a strategic framework used by companies to manage and control inventories of physical products, encompassing decisions related to setup, production, holding, ordering and shortage costs.

Stock-out - A situation in which there are no goods of a particular kind available for sale.

Holding cost - The cost of storing goods or owning assets.

Lead time - The time between ordering a product and receiving it.

Predictive analysis – This is the practice of using statistical algorithms and machine learning techniques to analyse historical data, identify patterns, and predict future outcomes.

Prescriptive analysis – This is the area of Business Analytics dedicated to searching out the best solution for day-to-day occurring problems.

Bulk to Break – This is the process of receiving large volume shipments of a single commodity and dividing them into smaller shipments or units of measure to be delivered to various customers or used in further production.

Unit of Measure – This is a definite magnitude of a quantity, defined and adopted by convention or by law, that is used as a standard for measurement of the same kind of quantity

Manual - refers to relying primarily on human judgment and fixed procedural rules rather than mathematically optimized decision models.

CHAPTER 1: INTRODUCTION

1.1. Background

Inventory is essential for any business's operation therefore proper inventory management is crucial for smooth running of day to day activities. Poor inventory management often leads to overstocking, shortages, tied-up working capital and increased operational demands. To mitigate these issues, manufacturing and distribution organizations worldwide are turning to data driven inventory control strategies that enable consistent and cost effective decision making. In particular, the food, retail and packaging supply industries are moving toward lean and evidence-based inventory practices that aim to minimize waste while maintaining service levels (Chopra & Meindl 2019).

The global landscape of operations research has been reshaped by the integration of predictive and prescriptive analytics. Contemporary inventory systems now harness Machine Learning (ML) to move beyond reliance on historical averages, applying supervised learning techniques to uncover intricate, non-linear demand behaviours and seasonal fluctuations (Zhang & Huang, 2020; Kumar & Goyal, 2025). These advanced forecasts then serve as essential inputs for prescriptive optimization approaches, including Linear Programming (LP) and Mixed-Integer Linear Programming (MILP) (Yousefi & Lari, 2022). In more developed economies, linear programming (LP) and mixed-integer linear programming (MILP) are frequently used to assist prescriptive inventory choices, enabling businesses to choose the best ordering schedules and replenishment quantities (Pinto & Ferreira, 2021; Vicente, 2025). These methods replace reactive, judgment-based inventory management techniques with organized, statistically based decision frameworks (Nguyen & Shiraki, 2021; Munyoka & Doyo, 2022).

Locally, many small-to-medium enterprises (SMEs), including packaging material suppliers, continue to rely on manual or heuristic-based inventory practices. Such businesses typically procure packaging materials such as carrier bags, packaging paper, bubble wrap, and adhesive tapes in bulk units such as rolls or kilograms, while sales are conducted in variable units including meters, pieces, or smaller bundles. This mismatch in units of measure, combined with demand variability, often leads to inaccuracies in stock records and increased time spent on periodic inventory reconciliation, particularly at the end of each month.

The client's challenge originates at the crossroads of demand volatility and unit conversion complexity. In the absence of predictive tools such as Machine Learning to forecast fragmented demand, and prescriptive models like MILP to systematically handle bulk conversions, the organization remains locked in a reactive mode of operation. This cycle culminates in a persistent reconciliation bottleneck, where staff are compelled to work long hours at the end of the month conducting manual stock counts and adjustments. The motivation for this research is the pressing need to mitigate these labour intensive inefficiencies by designing a hybrid ML–MILP framework. By establishing a unified source of truth through integrated forecasting and optimized conversion, the study seeks to eliminate the root causes of the end of month crisis and align the client's processes with contemporary, lean and mathematically robust practices.

1.2. Project Overview

This project is within the research area of prescriptive analytics and inventory optimization, emphasizing the integration of predictive modelling with mathematical programming to support evidence based replenishment strategies. Across the globe, operations research has advanced to help organizations navigate increasingly complex supply chains by leveraging computational methods that account for multi-dimensional constraints such as fluctuating demand, budgetary restrictions and capacity limits (Chopra et al., 2019; Zhang & Huang, 2020). This research departs from static inventory models by adopting a dual-stage Forecast then Optimize pipeline, a paradigm in which predictive outputs directly shape the parameters and constraints of optimization frameworks (Sani et al., 2021; Yousefi & Lari, 2022).

Machine learning techniques have been extensively applied to demand forecasting, leveraging historical sales data to uncover complex patterns and temporal dependencies (Zhang & Huang, 2020; Kumar & Goyal, 2025). While ML models excel at prediction, they do not inherently determine optimal replenishment decisions or account for rigid operational constraints such as storage capacity, fixed ordering costs and unit of measure conversions (Bertsimas & Kallus, 2018; Sani et al., 2021). In the packaging sector, where sales are fragmented and highly variable, forecasts alone cannot resolve the fundamental procurement challenge (Nguyen & Shiraki, 2021; Munyoka & Doyo, 2022). Accordingly, this research employs ML as a predictive layer to generate high-fidelity demand forecasts, capturing seasonal fluctuations and latent trends that inform subsequent decision making (Yousefi & Lari, 2022; Namazian et al., 2025).

To overcome the limitations of purely predictive models, prescriptive analytics, particularly Mixed-Integer Linear Programming (MILP), is utilized to translate predictive insights into actionable inventory decisions (Sani et al., 2021). MILP enables the simultaneous representation of discrete choices e.g., whether to place an order and continuous variables e.g., order quantities within a unified optimization framework (Pinto & Ferreira, 2021; Vicente, 2025). This mathematical rigor is critical for inventory systems that incur fixed delivery costs and must operate under capacity constraints (Zhang & Huang, 2020; Yousefi & Lari, 2022). Within this study, the MILP model functions as the computational engine that performs Unit of Measure conversions, effectively transforming fragmented demand forecasts from the ML stage into bulk raw material requirements. In doing so, procurement is precisely aligned with anticipated production needs, ensuring both efficiency and accuracy in inventory management.

At the local level, packaging material suppliers face difficulties such as inconsistent units of measurement, reliance on manual inventory tracking and time-consuming reconciliation tasks. To tackle these challenges this research seeks to establish a reliable, automated decision support framework. The proposed approach is intended not only to stabilize inventory levels but also to illustrate how advanced computational methods can be scaled to strengthen the resilience and efficiency of SME level supply chain operations. In doing so, it aims to reduce dependence on manual, error prone stock keeping practices and promote more consistent, data driven inventory management.

1.3.Statement of the Problem

Despite the critical role of inventory in operational efficiency, many packaging material suppliers continue to manage stock using manual checks, fixed interval ordering and subjective judgment. These heuristic based methods frequently overlook non-linear demand fluctuations, purchase lead times and the inherent mathematical discrepancies resulting from roll based inventory measurements (Nguyen & Shiraki, 2021; Munyoka & Doyo, 2022). A primary driver of this inefficiency is the Bulk to Break Unit of Measure mismatch, where raw materials are procured in bulk but sold in fragmented pieces or meters. As a result, businesses frequently encounter disparities between recorded and actual stock levels.

A major consequence of these practices is the recurring end of month inventory reconciliation procedure, which requires workers to put in extra time to confirm stock quantities, update records and make accurate purchase decisions. This not only raises the risk of human error but also interferes with regular operations and increases labour cost. Moreover, reactive inventory decisions also raise the risk of stock-outs, which result in lower service levels and lost sales or overstocking, which results in needless storage expenses.

A significant shortcoming in existing inventory management procedure is the absence of mathematically grounded decision support framework that integrates predictive and prescriptive intelligence. This deficiency creates a predictive gap in anticipating fragmented demand and a prescriptive gap in optimizing replenishment under complex Unit of Measure constraints (Nguyen & Shiraki, 2021). There is limited empirical evaluation of how optimization based inventory policies can reduce operational waste and stabilize inventory levels in packaging material supply environments (Munyoka & Doyo, 2022). This study therefore seeks to address this challenge by evaluating whether a hybrid framework that combines Machine Learning (ML) for demand forecasting with Mixed-Integer Linear Programming (MILP) for replenishment optimization can outperform manual heuristics. By minimizing total inventory costs and eliminating the sources of labour intensive reconciliation, the research aims to shift the organization from reactive stock management toward a proactive, lean operational model (Pinto & Ferreira, 2021; Vicente, 2025).

1.4.Proposed Solution

This research proposes an integrated ML–MILP prescriptive framework for optimizing inventory replenishment in packaging operations. The solution adopts a Forecast then Optimize pipeline: the first stage employs a Machine Learning (ML) predictive model to generate demand forecasts for high variance, fragmented sales patterns. These forecasts serve as stochastic inputs to the second stage, a Mixed-Integer Linear Programming (MILP) optimization engine. The MILP model determines reorder quantities and timing while explicitly incorporating holding costs, fixed ordering fees and the complex Unit of Measure conversion logic required to translate fragmented sales forecasts into bulk raw material requirements.

The proposed solution is strictly analytical, emphasizing the evaluation of the hybrid policy through controlled simulation experiments rather than commercial system deployment. Model performance will be benchmarked against existing manual, fixed-interval heuristics using metrics such as total operational cost, inventory stability, and stock count accuracy. By formalizing inventory decisions within a mathematically defined optimization framework,

this research seeks to substitute labour intensive and reactive stock adjustments with a systematic, evidence driven decision model designed to minimize inefficiencies linked to periodic inventory reconciliation. This shift from manual control to algorithmic management enables inventory levels to align with real-time demand patterns instead of relying on subjective projections.

1.5.Objectives

1.5.1. General Objective

To develop and evaluate an integrated Machine Learning (ML) and Mixed-Integer Linear Programming (MILP) framework for optimizing inventory replenishment and minimizing operational inefficiencies in packaging material supply operations.

1.5.2. Specific Objectives

1. To examine manual inventory replenishment practices used by packaging material suppliers by identifying ordering rules, reconciliation frequency and unit of measure handling.
2. To develop a Machine Learning predictive model to forecast high-variance demand for fragmented packaging products using historical sales data.
3. To formulate a Mixed-Integer Linear Programming (MILP) model that utilizes ML-generated forecasts to optimize bulk replenishment quantities and schedules under cost and Unit of Measure constraints.
4. To benchmark the performance of the integrated ML-MILP framework against manual heuristic policies through simulation to evaluate cost-efficiency and inventory stability.

1.6.Research Questions

1. What inventory replenishment practices, reconciliation procedures and Unit of Measure handling methods characterize manual inventory management among packaging material suppliers?
2. How effectively can Machine Learning algorithms predict non-linear demand patterns for fragmented packaging products?
3. How can an MILP model be formulated to integrate ML-driven demand forecasts with bulk to break conversion logic and operational constraints?
4. To what extent the integrated ML-MILP framework improve total inventory cost, inventory level stability and stock-count accuracy relative to current manual practices?

1.7.Justification

The rationale for this research stems from the ongoing inefficiencies in inventory management among small and medium sized enterprises, especially those dependent on manual and heuristic decision processes. These approaches frequently lead to irregular replenishment, inaccurate inventory records and labour intensive reconciliation activities. By applying a hybrid framework that integrates Machine Learning (ML) with a mathematically structured optimization framework, the study seeks to bridge a practical operational gap while promoting evidence based practices in inventory control.

From an academic perspective the study contributes to the field of Data Science and operations research by demonstrating the practical integration of predictive and prescriptive analytics. Specifically, it shows how machine learning based demand forecasting can be effectively combined with Mixed-Integer Linear Programming (MILP) to address Unit of Measure complexity in inventory systems. It further enriches the inventory optimization literature by situating MILP-based decision frameworks within SME supply contexts that remain insufficiently explored in empirical research.

From a practical standpoint, the proposed solution offers business owners and operations managers a systematic substitute for reactive inventory practices. The results are anticipated to enhance decision consistency, minimize operational inefficiencies and lessen dependence on labour intensive manual adjustments. In addition, the study reflects ISO 9001:2015 quality management standards by fostering data driven decision making and continuous improvement in processes.

1.8. Proposed System Methodologies

This research adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology, adapted to suit a prescriptive analytics research lifecycle. The choice of CRISP-DM is justified by its iterative structure and its emphasis on aligning mathematical models with organizational objectives, ensuring that the final hybrid framework addresses the distinctive Bulk to Break challenges of the packaging sector.

The lifecycle begins with data understanding and preparation, where the Pandas library is employed to standardize historical sales records and reconcile Bulk to Break unit inconsistencies. This foundational phase ensures that subsequent models are trained on high integrity datasets that accurately capture the fragmented demand patterns characteristic of the packaging sector.

The implementation stage follows a Forecast then Optimize pipeline. Predictive modelling is conducted using XGBoost (Extreme Gradient Boosting), selected for its ability to manage high variance, non-linear demand spikes through sequential boosting and L2 regularization, thereby mitigating the risk of overfitting in noisy SME datasets. The resulting forecasts are then ingested by a Mixed-Integer Linear Programming (MILP) model, formulated via the PuLP library in Python, to determine optimal replenishment schedules. This dual tool approach is justified because it enables the system to address complex operational constraints such as storage limitations and unit of measure conversions that cannot be resolved by predictive models alone.

The research concludes with a simulation based evaluation designed to benchmark the hybrid ML–MILP framework against existing manual heuristics. Controlled discrete event simulations will be used to validate performance, employing metrics such as total operational cost reduction, inventory level stability and stock-count accuracy. This quantitative approach ensures the study remains an analytical evaluation of mathematical policies, providing empirical evidence for reducing the labour intensive reconciliation cycles that currently hinder organizational efficiency.

1.9.Scope

The scope of this research is restricted to inventory replenishment decision making for packaging materials within a single operational setting. It concentrates on the interaction between demand forecasting (ML) and order optimization (MILP), specifically identifying optimal order quantities and timing, without extending to broader supply chain activities such as supplier selection, transportation, warehouse optimization or distribution management.

Additionally, the study is limited to assessing a mathematical optimization model through simulation rather than implementing a real time system or enterprise wide solution.

Constraints related to data availability, demand forecasting accuracy and parameter assumptions are recognized. The results will be interpreted within the specific operational and methodological boundaries established for the study.

CHAPTER 2: LITERATURE REVIEW

2.1. Introduction

This chapter examines the literature on inventory management and optimization, with a focus on data-driven decision models used in inventory replenishment and supply chain planning. It outlines the fundamental concepts and methodologies that inform the project, including traditional inventory control practices, optimization approaches such as Mixed-Integer Linear Programming (MILP) and machine learning based demand forecasting techniques.

The purpose of this review is to establish the theoretical foundation for the research and to identify gaps in current knowledge that this study seeks to address. In particular, this chapter examines the evolution of predictive analytics (machine learning) and prescriptive analytics (mathematical optimization) as complementary approaches to enhancing inventory decision making. It assesses the ability of these approaches to enhance demand estimation, decision quality and operational performance when compared to heuristic or rule based approaches. By situating the study within this body of knowledge, the chapter underscores both the practical relevance and academic contribution of adopting hybrid ML informed MILP frameworks in modern inventory management.

2.2 Theoretical Review

2.2.1 Inventory Management Concepts

Inventory management involves the planning and control of raw materials, work in progress and finished goods to ensure product availability while minimizing overall operational costs. Core variables frequently addressed in optimization literature include demand, replenishment quantity, ordering frequency, holding cost, ordering cost and stock out penalties (Zhang & Huang, 2020).

Demand represents the quantity of items required by customers within a given period and may be deterministic or stochastic. In small to medium enterprises (SMEs), demand is typically uncertain and influenced by seasonal fluctuations and customer specific factors (Nguyen & Shiraki, 2021). Holding cost refers to expenses associated with storing inventory, including warehousing, handling and capital tie-up. Ordering cost encompasses fixed administrative and transportation expenses incurred whenever an order is placed, regardless of size. Stock-out costs capture the losses incurred when inventory fails to meet demand, including lost sales and diminished customer satisfaction (Zhang, X., & Huang, Y. 2020).

In packaging material supply businesses, inventory management is further complicated by unit of measure variability. Materials such as bubble wrap, carrier bags, and packaging paper are often procured in bulk units like rolls or kilograms but sold in variable units such as meters or pieces. These complexities also motivate the need for methods that can predict future demand accurately and translate those predictions into optimized inventory decisions.

2.2.2 Inventory Control Policies

Inventory control policies establish rules for replenishment decisions. Traditional approaches include Economic Order Quantity (EOQ), periodic review and reorder point systems. While the Economic Order Quantity (EOQ) provides a foundational framework for cost minimization, its practical utility is constrained by assumptions of constant demand and the absence of storage capacity or integer constraints that do not hold under real operational

conditions limiting its applicability in realistic, variable environments (Essien, E.E.& Otu, U.O.E. 2024). Heuristic and rule based policies remain common in SMEs due to their simplicity. These rely on judgment, fixed reorder intervals or visual stock checks. While straightforward to implement, such methods often fail to minimize costs under demand variability and operational constraints (Nguyen & Shiraki, 2021).

Optimization-based policies, particularly those employing Linear Programming (LP) and MILP, provide a prescriptive alternative. MILP models incorporate discrete decision variables, fixed ordering costs, storage capacity constraints, and multiple time periods, making them well suited to real world inventory systems (Pinto & Ferreira, 2021).

2.2.3 Optimization and MILP in Inventory Decision Making

Optimization techniques address decision problems by identifying the best possible outcome according to a defined objective function subject to constraints. In inventory management, the objective function typically seeks to minimize total costs including holding, ordering, and stock-out penalties while ensuring demand satisfaction and adherence to operational limits. Mixed-Integer Linear Programming (MILP) extends traditional linear programming by incorporating integer and binary variables, thereby enabling the modelling of discrete decisions such as whether to place an order and how to account for capacity restrictions. Research on MILP models for inventory and supply chain planning illustrates how discrete scheduling and ordering decisions can be encoded in mathematical constraints to minimize total costs in multi-period settings (Vicente, J. J. 2025).

MILP provides a prescriptive analytics framework that translates demand forecasts into actionable replenishment decisions by solving a structured optimization problem. Its key strengths include: Representation of both continuous decisions e.g., order quantities and discrete decisions e.g., order placement, Integration of operational constraints such as storage capacity and fixed ordering costs. Provision of optimal solutions under defined cost parameters.

2.2.4 Why Hybrid ML + MILP (Predictive–Prescriptive Analytics)

Predictive analytics which involves forecasting based on data and prescriptive analytics which involves optimization of decisions represent two complementary stages of data driven decision making. A predictive system focuses on anticipating future demand, while a prescriptive system determines the optimal response to that demand. Integrating these two approaches, commonly referred to as predictive–prescriptive analytics, is increasingly recognized in operations research as a robust framework for decision support (Bertsimas & Kallus, 2018).

Recent literature highlights the effectiveness of hybrid models in supply chain and inventory contexts. For instance, hybrid frameworks that combine demand forecasting with inventory decision optimization have been shown to improve performance relative to traditional policies, particularly under stochastic demand conditions. For instance, predict and optimize models consistently achieve lower total costs than periodic review approaches (Namazian, Z., Betts, J.M. & Stuckey, P.J 2025). Similarly, applied studies demonstrate that AI based forecasting followed by LP or MILP optimization improves service levels and reduces costs compared to traditional heuristic methods (Kumar & Goyal, 2025).

Integrating ML and MILP enables systems to employ machine learning to generate accurate demand forecast and apply MILP optimization to determine optimal order quantities and timing under cost and operational constraints. This hybrid architecture reflects current industry trends toward data driven, automated decision systems, positioning predictive–prescriptive analytics as an emerging technology in supply chain and inventory optimization.

2.3 Case Study Review

Empirical applications of inventory optimization models highlight their relevance in operational contexts, particularly SMEs and supply oriented businesses. Nguyen and Shiraki (2021) found that SMEs often rely on manual heuristics despite frequent stock-outs and excess inventory. Their study showed that optimization-based policies significantly reduced inventory costs compared to fixed-interval replenishment methods. Simulation-optimization approaches have also been explored. Yousefi and Lari (2022) integrated MILP decision rules with stochastic demand scenarios, demonstrating improved inventory stability and cost reduction relative to heuristics, though implementation complexity remained a limitation

Research focusing on African small enterprises further highlights limited adoption of mathematical decision-support tools. Munyoka and Doyo (2022) examined African small enterprises and identified challenges including inconsistent record keeping, demand uncertainty and unit of measure discrepancies, conditions that closely mirror packaging material suppliers. These constraints often discourage the use of structured optimization despite their potential benefits.

Recent studies highlight the importance of integrating predictive and prescriptive analytics. Hybrid methodologies that combine machine learning driven demand forecasting with optimization techniques have demonstrated greater adaptability to dynamic demand fluctuations, while simultaneously ensuring cost-effectiveness and adherence to operational constraints (Kumar & Goyal, 2025). Findings suggest that although MILP models are effective in producing optimal solutions, the inclusion of enhanced demand forecasts significantly improves overall system efficiency, particularly in contexts where demand is influenced by nonlinear trends or external factors.

Most existing case studies predominantly examine manufacturing or large scale retail supply chains. In contrast, small and medium-sized enterprises (SMEs) engaged in packaging material distribution, particularly those managing mixed procurement and sales formats such as rolls versus individual pieces and relying on manual stock reconciliation, have received comparatively little empirical investigation. This scarcity of context specific evidence underscores a research gap that the present work aims to fill by assessing a hybrid inventory framework that integrates machine learning based forecasting with MILP optimization in the SME packaging sector.

2.4 Integration and Architecture

Inventory optimization models are typically deployed as decision support tools rather than full transactional systems. In practice, MILP based systems are implemented using modular architectures comprising data pre-processing, optimization and performance evaluation components (Pinto & Ferreira, 2021). Historical sales and inventory data are standardized into parameters, after which the optimization model computes replenishment decisions. These are then evaluated using cost and stability metrics.

For SMEs, lightweight implementations of MILP are preferred. Python libraries such as PuLP support the formulation and solution of mixed-integer optimization problems with open-source solvers, enabling developers to create discrete decision models for inventory replenishment, capacity limits and ordering decisions without requiring large enterprise systems. This supports simulation-based analysis while remaining accessible to resource-constrained businesses.

The proposed hybrid ML–MILP framework advances existing architectures by coupling a predictive component with a prescriptive optimization engine. Historical records of sales and inventory are employed to train and validate forecasting models such as machine learning or time series approaches yielding demand projections that feed into the MILP optimization process. The MILP module subsequently determines cost-minimizing inventory strategies while adhering to operational requirements, including storage limitations, fixed ordering costs, and unit-of-measure conversions.

This end to end decision-support pipeline: forecasting module → optimization module → performance evaluation, constitutes a cutting-edge prescriptive analytics paradigm for data driven inventory management (Bertsimas & Kallus, 2018). By uniting predictive forecasting with optimization, the framework harnesses advanced computational methods to improve decision accuracy, mitigate inefficiencies, and equip SMEs with practical, evidence-based inventory policies.

2.5 Summary

Existing research demonstrates that inventory management is strengthened through the integration of predictive and prescriptive techniques. Machine learning contributes to improved forecasting precision, while MILP delivers optimal decisions that account for operational constraints. The convergence of these methods produces a comprehensive predictive–prescriptive analytics framework, which is increasingly acknowledged as a leading approach for advancing data-driven inventory management systems.

2.6 Research Gaps

Although interest in inventory optimization and hybrid decision-making frameworks has grown, notable research gaps persist. Firstly, few empirical investigations have explored the deployment of ML enhanced MILP models within small to medium packaging material suppliers, particularly in the context of developing economies. Second, the effectiveness of hybrid approaches in inventory environments marked by unit of measure discrepancies, where procurement and sales units differ, remains insufficiently examined. Third, comparative analyses evaluating the performance of ML–MILP hybrids against conventional heuristic or standalone optimization techniques in SME inventory settings are limited.

To address these gaps, the present study proposes the design and evaluation of a predictive–prescriptive inventory optimization framework that integrates machine learning based forecasting with MILP optimization, specifically tailored to a packaging material supplier operating with mixed units of measure. Through simulation driven experimentation, the research aims to demonstrate the practical benefits of combining forecasting and optimization in enhancing inventory stability, lowering operational costs and strengthening decision-making within SME operations.

References

- Pinto, L. M., & Ferreira, R. J. (2021). Mixed-Integer Linear Programming for inventory and production planning. *International Journal of Production Economics*.
- Namazian, Z., Betts, J.M. & Stuckey, P.J. (2025). Predict and optimize: a smart inventory management model for beauty retail. *Optim Lett*
- Munyoka, W., & Doyo, N. (2022). Inventory practices and performance in small enterprises in Africa. *African Journal of Economic and Management Studies*.
- Nguyen, T. T. H., & Shiraki, Y. (2021). Inventory policies in SMEs under demand variability: A case analysis. *Journal of Small Business Management*.
- Pinto, L. M., & Ferreira, R. J. (2021). Mixed-Integer Linear Programming for inventory and production planning. *International Journal of Production Economic*
- Yousefi, S., & Lari, A. (2022). Simulation-optimization frameworks for inventory control. *Simulation Modelling Practice and Theory*.
- Zhang, X., & Huang, Y. (2020). Inventory optimization under demand uncertainty: A review of methods. *European Journal of Operational Research*.
- Sani, A., Bertsimas, D., & Dunn, J. (2021). Predict to prescribe: An integrated decision-making framework. *INFORMS Journal on Computing*
- Kumar, V., & Goyal, S.(2025). AI-Driven Forecasting and Optimization for Inventory Control in Manufacturing Supply Chain. *ACR Journal*.
- Vicente, J. J. (2025). Optimizing supply chain inventory: A mixed integer linear programming approach. *Systems*, 13(1), 33.
- Bertsimas, D., & Kallus, N. (2018). From predictive to prescriptive analytics. *arXiv*
- Chopra, S., Meindl, P., & Kalra, D. V. (2019). *Supply Chain Management: Strategy, Planning, and Operation* (7th ed.). Pearson Education.
- Essien, E. E., & Otu, U. O. E. (2024). An assessment of economic order quantity and organizational performance: A literature review. *Business & Social Sciences Journal (BS SJ)*.

APPENDICES

APPENDIX A: PROJECT RESOURCES

This appendix outlines the key resources required to successfully conduct the research project.

Hardware Resources

Personal computer or laptop.

Internet connectivity for literature access and software installation.

Software Resources

Python Programming Language for data analysis, machine learning, and optimization

Python Libraries:

- Pandas – data pre-processing and analysis
- NumPy – numerical computations
- Scikit-learn – machine learning demand forecasting
- PuLP – Mixed-Integer Linear Programming modeling
- Matplotlib / Seaborn – visualization of results

VS Code – development environment

Microsoft Excel – data entry, cleaning, and preliminary analysis

Microsoft Word – documentation and report writing

Data Resources

Historical sales data (quantities sold per period)

Inventory records (opening stock, closing stock, reorder quantities)

Cost parameters:

- Holding cost
- Ordering cost
- Stock-out or shortage cost

Unit of measure conversion information (e.g., roll-to-meter or kg-to-piece conversion)

Human Resources

Researcher (student)

Business owner or manager for operational insights and validation.

Academic supervisor (guidance and evaluation)

APPENDIX B: PROJECT BUDGET

Item	Description	Estimated Cost (KES)
Data Collection	Communication, printing, and transport	5,000
Internet Access	Literature access and software resources	3,000
Software Tools	Open-source tools (Python, PuLP, Scikit-learn)	0
Documentation	Printing and binding of final report	2,500
Contingency	Unforeseen project expenses	1,500
Total Estimated Cost		12,000

Table1: Budget

APPENDIX C: PROJECT SCHEDULE

Weeks	Activity Description
Weeks 1–2	System scoping, problem formalization, and data familiarization
Weeks 3–4	Data pre-processing, exploratory analysis, and demand characterization
Weeks 5–6	Development and validation of demand forecasting model
Weeks 7–8	Formulation and implementation of the MILP inventory optimization model
Weeks 9–10	Integration of forecasting outputs with optimization model and simulation setup
Week 11	Performance evaluation and comparison with manual heuristic policies
Week 12	Results interpretation, documentation, and final report preparation

Table 2: Schedule

APPENDIX D: GANTT CHART

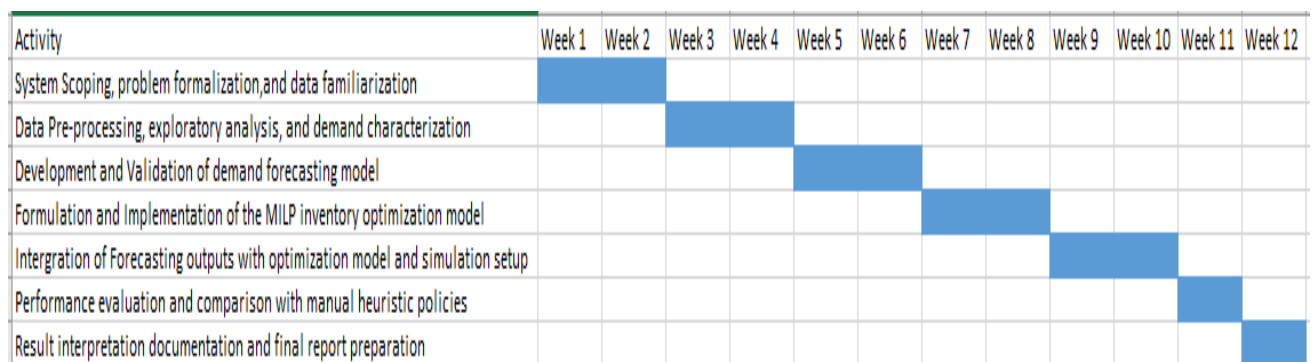


FIGURE 1